BEU CONNECT

Aryabhatta Knowledge University, Patna

B.Tech. 3rd Semester Examination, 2015

Time: 03 Hours

Code: 021305 ***Subject: Material Science*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer (any seven): **[14]**

(a) Which of the following materials is the hardest?
(i) Alumina (iii) Cast iron
(ii) High-carbon steel (iv) Diamond

(b) The phase boundary between α and (α+β) regions is called
(i) liquidus (iii) solvus
(ii) solidus (iv) eutectic

(c) The interstitial solid solution of carbon in γ-iron is called
(i) cementite (iii) austentite
(ii) ferrite (iv) pearlite

(d) For a spherical particle of radius r, the volume to surface area ratio is
(i) $r/3$ (iii) $\pi r/3$
(ii) $3r$ (iv) None of these

(e) Which of the following refers to eutectoid reaction?
(i) $L \rightarrow \alpha+\beta$ (iii) $\alpha + L \rightarrow \beta$
(ii) $\alpha \rightarrow \beta+\gamma$ (iv) $\alpha+\beta \rightarrow \gamma$

(f) If one solid phase splits into two solid phases on heating, the reaction is
(i) eutectoid (iii) eutectic
(ii) peritectoid (iv) peritectic

(g) Cast iron normally contains
(i) 1 to 1.5% C (iii) less than 1% C
(ii) 2 to 4% C (iv) None of the above

(h) Piezo-electric materials produce electric field when subjected to
(i) magnetic field (iii) thermal radiation
(ii) optical rays (iv) mechanical force

(i) Examples of ferrous metals are
(i) copper, brass, nickel (iii) cast iron, steel, alloy steel
(ii) aluminium, titanium, zinc (iv) alumina, zirconia, silica

(j) In a single-component system, the maximum no. of phases that can co-exist in equilibrium, is
(i) 2 (iii) 4
(ii) 3 (iv) 5

Q.2 Solve these questions: **[14]**

(a) State and explain the equation for Gibbs phase rule.

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store

(b) Derive the lever rule for the amount in wt. percent of each phase in two phase regions of a binary phase diagram.

Q.3 Solve these questions: **[14]**

(a) What is a hybrid composite?

(b) Write the important advantages of hybrid composites over normal fiber composites.

(c) Find an expression for the modulus of elasticity for a hybrid composite in which all fibers of both types are oriented in the same direction.

Q.4 Solve both questions : **[14]**

(a) Explain with reason why glass fiber reinforced composites are utilized extensively.

(b) Explain the limitations of glass fiber reinforced composites.

Q.5 Solve this question: **[14]**

Compare gray and malleable cast iron with respect to:

(a) composition and heat treatment;

(b) microstructure;

(c) mechanical characteristics.

Q.6 Solve these questions: **[14]**

(a) What is the distinction between cement and concrete?

(b) Write some important limitations that restrict the use of concrete as structural material.

(c) Explain the techniques that are utilized to strengthen concrete by reinforcement.

Q.7 Solve both questions : **[14]**

(a) Classify different types of materials with their properties and application.

(b) Write short notes on the following: (i) Biomaterial (ii) Optical fiber (iii) Nanomaterials

Q.8 Solve these questions: **[14]**

(a) Draw the lead-tin phase diagram.

(b) What are the compositions of phases?

(c) Calculate the relative amount of each phase present in terms of mass fraction.

Q.9 Write notes on the following: **[14]**

(a) (a) Bainite

(b) (b) White cast iron

(c) (c) Martensite

(d) (d) Hume-Rothery Rules

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store